Medical Diagnosis Expert System Using Logic Programming and Rule-Based Knowledge Representation

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***Abstract*—This project presents a rule-based medical diagnosis expert system designed to assist healthcare professionals and students in preliminary diagnostic reasoning. The system collects patient symptoms and selected medical-history information, represents them as structured facts, and applies explicit inference rules to suggest a possible disease with supporting evidence. The prototype includes both a console-based diagnosis engine and a Flask-based web interface. It also supports disease-specific treatment information and PDF report generation. The system currently covers 22 diseases, including respiratory, gastrointestinal, metabolic, vector-borne, cardiovascular, eye-related, and general conditions. The main contribution of this project is the implementation of an interpretable expert system where diagnostic conclusions can be traced back to symptoms and rules. The system is intended for educational and decision-support purposes only and is not a substitute for professional medical diagnosis.**

***Index Terms*—Expert system, medical diagnosis, rule-based inference, knowledge representation, clinical decision support, Flask, Experta.**

I. Introduction

A. Motivation

Medical diagnosis is a complex reasoning process that requires the interpretation of symptoms, medical history, risk factors, and clinical knowledge. In real healthcare practice, this work is performed by trained professionals. However, expert systems can support this reasoning process by organizing medical knowledge into structured rules and applying those rules consistently. The motivation behind this project is to demonstrate how artificial intelligence techniques, especially logic programming and knowledge representation, can be used to build a transparent medical diagnosis support system. Unlike black-box machine learning models, a rule-based expert system allows users to inspect why a particular diagnosis was suggested.

B. Project Objectives

The main objectives of this project are:

- To design a medical diagnosis expert system based on symptoms and medical history.
- To represent disease knowledge using facts, rules, and inference logic.
- To implement both console-based and web-based diagnosis interfaces.
- To provide treatment information for diagnosed diseases.
- To generate a simple report or PDF output for consultation results.
- To keep the system interpretable so that each diagnosis can be explained through matched symptoms.

C. Scope of the System

The system supports 22 diseases and asks users a series of symptom-related questions. Based on the answers, it applies rule-based reasoning to infer a possible disease. The system is educational and advisory. It does not perform laboratory analysis, imaging interpretation, physical examination, or emergency triage. Therefore, its output should be treated as a preliminary suggestion and reviewed by a healthcare professional.

II. Background and Related Work

A. Expert Systems

An expert system is an artificial intelligence program that uses domain knowledge and inference rules to solve problems that normally require human expertise. A typical expert system contains a knowledge base, an inference engine, and a user interface. The knowledge base stores facts and rules, while the inference engine applies those rules to reach conclusions. One of the most famous medical expert systems is MYCIN, which used rule-based reasoning to support diagnosis and

treatment of bacterial infections. MYCIN demonstrated that medical knowledge could be encoded as production rules and used for consultation.

B. Rule-Based Medical Diagnosis

Rule-based diagnosis systems use IF-THEN rules. For example: IF fever is high AND cough is present AND breathing difficulty is present THEN pneumonia may be suspected. This approach is useful because medical reasoning often depends on combinations of symptoms. A single symptom such as fever is not enough to identify a disease, but fever combined with rash, body pain, and bleeding tendency may support a dengue diagnosis.

C. Clinical Decision Support Systems

Clinical decision support systems help healthcare professionals by providing patient-specific recommendations, reminders, or diagnostic suggestions. According to AHRQ and WHO, such systems must be designed carefully, because healthcare decisions involve patient safety, privacy, and clinical responsibility. The proposed system follows this principle by presenting diagnoses as suggestions rather than final medical decisions.

III. System Design and Methodology

A. Overall Architecture

The system has five main components:

- User Interface
- Session and Question Controller
- Inference Engine
- Disease Knowledge Base
- Treatment and Report Module

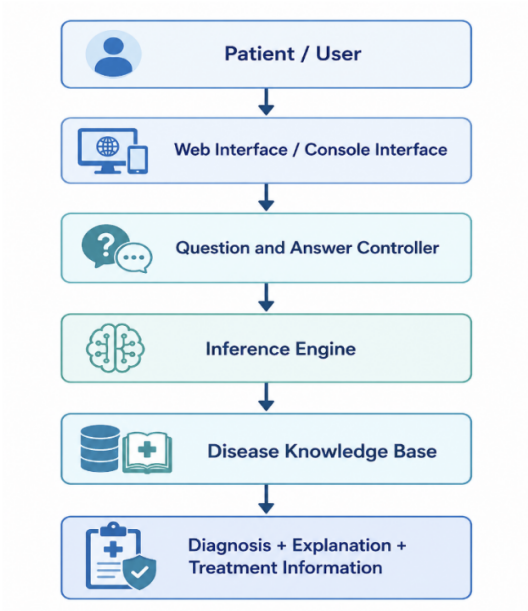

Fig. 1. Overall System Architecture

B. Knowledge Representation

The system represents user responses as facts. Examples include:

TABLE I
Knowledge Representation

Fact Type	Example
General symptom	fever type
Respiratory symptom	cough, breathing difficulty
Gastrointestinal symptom	vomiting, abdominal pain
Eye symptom	redness, itching
Medical history	allergy, medication history

A disease rule is represented as a combination of required or supporting symptoms. A simple evidence score can be written as:

$$\text{Score}(d) = \frac{|\text{MatchedSymptoms}(d)|}{|\text{RequiredSymptoms}(d)|}$$

C. Inference Mechanism

The system uses forward-style reasoning. It starts with user answers, converts them into facts, and checks which disease rules match those facts. If a rule condition is satisfied, the system returns a possible diagnosis and supporting symptoms.

Fig. 2. Inference Flow Diagram

D. Rule Design Strategy

The rules are designed to avoid diagnosis based on one generic symptom. For example, fever alone does not identify dengue, malaria, influenza, or pneumonia. Therefore, disease rules use combinations of symptoms.

Example:
IF high fever = yes
AND body pain = yes
AND rash = yes
AND bleeding tendency = yes
THEN possible diagnosis = Dengue

IV. Implementation

A. Development Tools and Technologies

The system was developed using the following tools:

TABLE II
Development Tools and Technologies

Tool/Technology	Purpose
Python	Main programming language
Experta	Rule-based expert system logic
Flask	Web application framework
HTML/CSS	Web interface design
Markdown/HTML treatment files	Disease information
PDF generation logic	Report download support

B. Console Diagnosis Engine

The console diagnosis engine is implemented in expert.py. It asks users questions, collects answers, applies rule-based logic, and prints the possible diagnosis. This version demonstrates the core expert-system reasoning without requiring a browser.

Fig. 3. Code screenshot of the rule-based diagnosis engine in expert.py.

C. Web-Based Interface

The web version is implemented in web.py using Flask. It provides a local browser-based interface where users answer diagnosis questions step by step. The system stores answers using session identifiers and sends responses to the diagnosis engine. Important web features include:

- Question-by-question consultation
- Session-based answer storage
- Diagnosis output
- Treatment page linking
- PDF/report download support

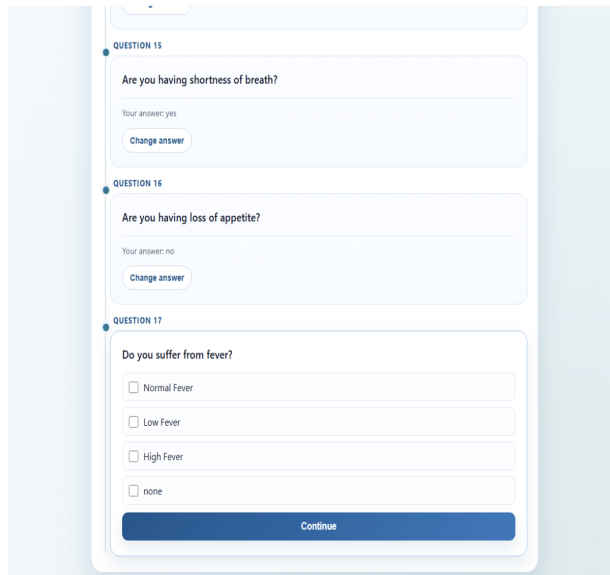

Fig. 4. Question answer interface for disease detection.

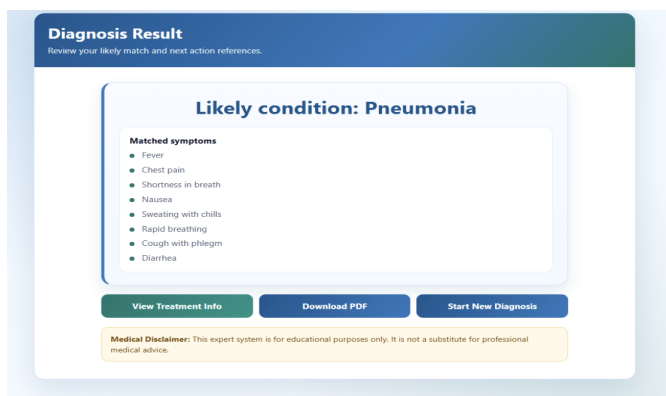

Fig. 5. Diagnosis result page showing possible disease and supporting evidence.

D. Treatment Information and PDF Support

After a diagnosis is generated, the system can show treatment-related information from the Treatment folder. The treatment files are stored separately from the rule engine, which makes the system easier to maintain. The web version also supports report/PDF generation so that consultation results can be saved.

V. Disease Knowledge Base

A. Supported Diseases

The system supports 22 diseases:

1) Diabetes
2) Coronary arteriosclerosis
3) Pneumonia
4) Asthma
5) Anemia
6) Hypothyroidism
7) AIDS/HIV infection
8) Dehydration
9) Obesity
10) Arthritis
11) Bronchitis
12) Pancreatitis
13) Peptic ulcer
14) Gastritis
15) Hepatitis
16) Influenza
17) Tuberculosis
18) Malaria
19) Dengue
20) Coronavirus disease
21) Conjunctivitis
22) Eye allergy

B. Symptom Categories

The symptoms can be grouped into broad categories:

TABLE III
DISEASE CATEGORY DISTRIBUTION

Category	Example Symptoms
Respiratory	cough, chest pain, breathing difficulty
Gastrointestinal	vomiting, abdominal pain, acidity
Vector-borne	fever, chills, rash, body pain
Metabolic	thirst, frequent urination, weight change
Eye-related	redness, itching, watery discharge
General	fatigue, weakness, fever

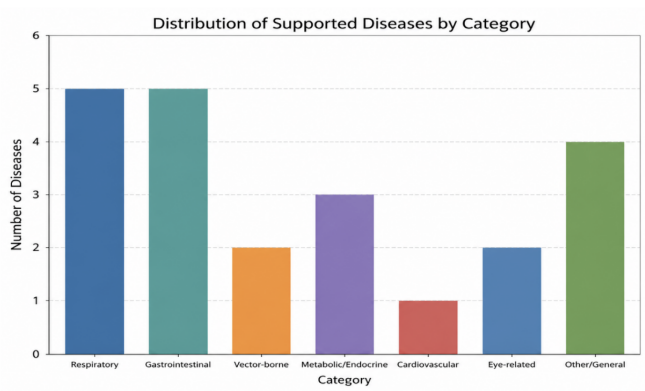

Fig. 6. Distribution of supported diseases by category.

C. Example Diagnosis Rules

Example respiratory rule: IF fever = high AND breathing difficulty = yes. THEN possible diagnosis = Pneumonia. Example metabolic rule: IF frequent urination = yes AND excessive thirst = yes AND weight loss = yes. THEN possible diagnosis = Diabetes Example eye-related rule: IF eye redness = yes AND itching = yes AND watery discharge = yes. THEN possible diagnosis = Eye Allergy

VI. TESTING AND EVALUATION

A. Functional Test Scenarios

The system was tested using scenario-based functional tests.

TABLE IV
FUNCTIONAL TEST SCENARIOS

Test Case	Input Pattern	Expected Output
Pneumonia case	fever, cough, breathing difficulty	Pneumonia
Dengue case	high fever, body pain, rash	Dengue
Diabetes case	thirst, urination, weight loss	Diabetes
Eye allergy case	itching, redness, watery discharge	Eye allergy

TABLE V
ADDITIONAL TEST SCENARIOS

Test Case	Input Pattern	Expected Output
Generic fever case	fever only	No overconfident diagnosis
Missing answer case	incomplete response	Ask next question
Treatment page case	known disease	Correct treatment page

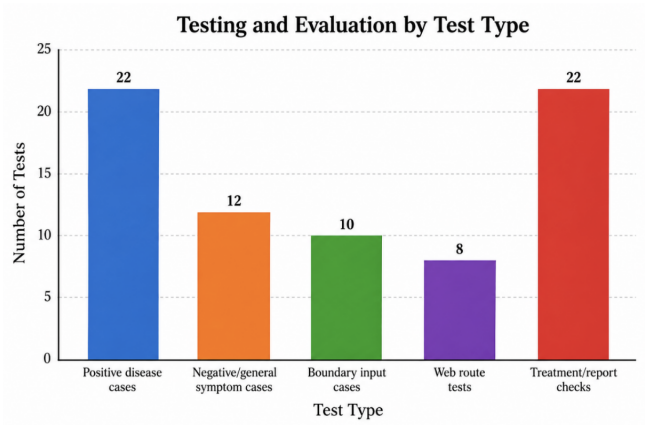

Fig. 7. Scenario-based testing coverage of the diagnosis system.

B. Sample Consultation Trace

System: Do you have fever? User: Yes System: What type of fever? User: High fever System: Do you have cough? User: Yes System: Do you have breathing difficulty? User: Yes System: Do you have chest pain? User: Yes Diagnosis: You may be suffering from Pneumonia. Evidence: high fever, cough, breathing difficulty, chest pain. This trace shows how the system collects symptoms and uses them as evidence for diagnosis.

C. Evaluation Limitations

The testing confirms that the software logic works for expected cases. However, it does not prove clinical accuracy. Real medical validation would require expert-reviewed patient cases, comparison with physician diagnosis, and clinical performance metrics such as sensitivity and specificity.

VII. Discussion

A. Strengths of the System

The main strengths of the system are:

- The reasoning process is transparent.
- Rules are easy to inspect and modify.
- The system supports both console and web interfaces.
- Disease-specific treatment information is connected to diagnosis output.
- The system demonstrates core expert-system concepts clearly.

B. Limitations

The system has several limitations:

- It uses manually written rules.
- It does not use laboratory results or imaging data.
- It does not consider disease probability or prevalence.
- It cannot replace clinical examination.
- Some diseases share similar symptoms, which may reduce certainty.
- The knowledge base requires expert review before real healthcare use.

C. Safety and Ethical Considerations

Medical diagnosis systems must be used carefully. Incorrect suggestions may create false confidence or unnecessary concern. Therefore, the system should clearly state that it is for educational and decision-support purposes only. Emergency symptoms such as severe chest pain, breathing difficulty, bleeding, or severe dehydration should always be handled by qualified healthcare professionals.

VIII. Future Work and Conclusion

A. Future Improvements

Future improvements may include:

- Storing rules in a database or JSON file.
- Adding rule source citations and review dates.
- Adding severity detection and red-flag warnings.
- Adding clinician review mode.
- Adding more diseases and symptoms.
- Improving the web interface.
- Adding multilingual support.
- Adding automated unit tests for every disease rule.
- Adding probabilistic ranking when multiple diseases match.

B. Conclusion

This project successfully demonstrates a medical diagnosis expert system using logic programming and rule-based knowledge representation. The system converts patient symptoms and history into facts, applies disease rules, and produces possible diagnoses with supporting evidence. It includes both console and web-based implementations, treatment information, and report support. Although the system is not clinically validated, it effectively demonstrates expert-system reasoning, explainability, and knowledge-based medical decision support in an educational context.

References

[1] B. G. Buchanan and E. H. Shortliffe, Eds., *Rule-Based Expert Systems: The MYCIN Experiments of the Stanford Heuristic Programming Project*. Reading, MA, USA: Addison-Wesley, 1984.

[2] E. H. Shortliffe, *Computer-Based Medical Consultations: MYCIN*. New York, NY, USA: Elsevier, 1976.

[3] E. H. Shortliffe and B. G. Buchanan, "A model of inexact reasoning in medicine," *Mathematical Biosciences*, vol. 23, no. 3–4, pp. 351–379, 1975.

[4] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2021.

[5] R. T. Sutton, D. Pincock, D. C. Baumgart, D. C. Sadowski, R. N. Fedorak, and K. I. Kroeker, "An overview of clinical decision support systems: Benefits, risks, and strategies for success," *npj Digital Medicine*, vol. 3, Art. no. 17, 2020, doi: 10.1038/s41746-020-0221-y.

[6] World Health Organization, *WHO Guideline: Recommendations on Digital Interventions for Health System Strengthening*. Geneva, Switzerland: WHO, 2019. [Online]. Available: https://www.who.int/publications/i/item/9789241550505

[7] Agency for Healthcare Research and Quality, "Clinical decision support," *Digital Healthcare Research*. [Online]. Available: https://digital.ahrq.gov/health-it-tools-and-resources/clinical-decision-support-cds

[8] Experta, "The basics — Experta documentation." [Online]. Available: https://experta.readthedocs.io/en/latest/thebasics.html

[9] Pallets Projects, "Flask documentation." [Online]. Available: https://flask.palletsprojects.com/